

Executive Overview

CTP Job Cards — Executive Overview

What the System Is

The CTP Job Cards platform is a real-time digital maintenance management system built for the plant. It replaces paper-based, radio, and verbal fault-reporting processes with a structured, tracked, and auditable system accessible from every employee's mobile phone.

The platform is built on Google Firebase — enterprise-grade cloud infrastructure — and is designed to be always available, offline-capable, and scalable without requiring any on-site IT infrastructure.

The Problem It Solves

In a traditional plant environment, fault reporting and maintenance response suffer from well-known problems:

- **No formal record** — faults reported verbally or on paper are frequently lost, incomplete, or undiscoverable after the event
- **No accountability** — there is no reliable way to know when a fault was reported, when it was attended to, or how long it took to resolve
- **No real-time visibility** — managers are often the last to know about active faults or unresolved jobs
- **No pattern recognition** — without structured records, it is impossible to identify which machines fail most often, at what cost, or why
- **Slow response** — critical faults rely on physically finding a technician or making a radio call, which introduces dangerous delays

The CTP Job Cards system addresses all of these by digitising the entire process — from fault report to job closure — with automated notifications, escalation, and a permanent audit trail.

The Four User Types

Role	Primary Function
Operator	Discovers a fault and creates a job card. The first link in the chain — quality of entry determines quality of the record.
Technician	Receives the job notification, attends to the fault, records what was done, and closes the job.
Manager	Oversees all jobs in their department, enforces data quality, monitors escalations, and identifies recurring failure patterns.
Admin	Manages system configuration, employee accounts, geofence settings, and access control.

How the System Works — High Level

1. Fault Reported

An **operator** identifies a fault and creates a job card on their phone. They record the machine, area, component, fault description, and priority (1 to 5, where 1 has no production impact and 5 means production is standing).

The job card is live the moment it is submitted.

2. Relevant Technicians Notified Instantly

The system automatically identifies which **technicians** are on site (via real-time geofencing) and what their trade is (mechanical, electrical, or both). The appropriate on-site technicians receive push notifications on their phones immediately — no dispatcher, no radio call.

Notification urgency scales with priority:

- **Priority 1–3:** Standard notification banner
- **Priority 4:** Persistent banner with action buttons that stays visible until responded to

- **Priority 5:** Full-screen alarm that overrides the lock screen and Do Not Disturb — cannot be passively ignored

3. Technician Accepts or Responds

The technician receives the alert and can assign themselves to the job, report themselves as busy, or dismiss it — directly from the notification. Every response is logged with a timestamp.

4. Automatic Escalation

Escalation runs across **four configurable stages**, with timers, recipient lists, and on/off switches managed by Admin under Settings → Escalation Rules. By default, if no technician responds within 5 minutes, on-site managers and foremen are notified; at 10 minutes the on-site department managers and workshop manager receive an urgent alert. Stages 3 (30 min) and 4 (60 min) are available but disabled by default, ready to be switched on if a site requires further escalation tiers. No critical fault can be silently ignored, and the timing can be tuned to match site conditions.

5. Job Tracked to Closure

The technician self-assigns from the alert (status moves automatically from **Open to In-Progress**), works the job, and updates its status as it progresses (Open → In-Progress → Monitor → Closed). When complete, they record what was done. The operator who raised the job is notified of the closure. The full record is stored permanently.

6. Management Oversight

Managers have a dedicated dashboard with live visibility of all department jobs, technician on-site status, escalation history, and notification logs. They are responsible for policing job card quality — ensuring all entries are complete and accurate.

Job Card Status Flow

Created → Open → In-Progress → Monitor → Closed

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(direct close)

Status	Meaning
Open	Job created — awaiting technician acceptance
In-Progress	Technician has self-assigned and is actively working the job
Monitor	Fault resolved, machine under observation
Closed	Job complete and confirmed resolved

What the System Captures

Every job card creates a permanent, searchable record:

Data Point	Business Value
Machine, area, part	Asset identification for tracking
Fault description	What happened and why
Priority at time of report	Urgency and production impact
Who created it, when	Operator accountability and fault timestamp
Who responded, when	Technician accountability and response time
What was done	Repair record and parts used
Time from report to closure	Mean time to repair (MTTR)
Full notification history	Who was notified, when, and what they did

Geofencing and On-Site Tracking

The system uses geofencing technology to automatically determine whether each employee is within the site boundary:

1. **Smart notification routing** — only on-site technicians receive job alerts, preventing off-hours disturbance for faults they cannot attend
2. **Site attendance records** — every entry and exit is logged with GPS coordinates and timestamp

Geofence events are triggered in real time using native Android location services, with a 30-minute background check for devices that may have missed an event.

Security and Access Control

The system uses **Firebase Authentication** — users log in with their company email and password, linked to their clock number. Three access levels exist:

Role	Access Scope
Operator / Technician	Own jobs, relevant department notifications
Manager	Full department visibility, escalation logs, notification history
Admin	Full system — employee management, geofence configuration, all data

All data is encrypted at rest and in transit. No employee can access data outside their authorised scope.

Infrastructure and Cost Model

Component	Provider	Notes
Database	Google Firebase Firestore	Real-time, offline-capable, serverless
Push Notifications	Firebase Cloud Messaging	Enterprise-grade, free tier supports high volumes
Cloud Functions	Google Cloud Functions	Serverless — no server to maintain
Authentication	Firebase Auth	Email + password, linked to clock numbers
Mobile App	Android (primary)	iOS and web capable

There is no server to maintain, no on-site hardware to manage, and no single point of failure. The system scales automatically.

Current Capabilities

Capability	Status
Real-time job card creation and tracking	Live
Push notifications scaled by priority (P1–P5)	Live
Full-screen critical alarms (P5)	Live
Automatic escalation (4 configurable stages; 2 enabled by default)	Live
Light and dark theme support	Live
Real-time geofencing and on-site status	Live
Background location check (30-min fallback)	Live
Permanent audit trail for every action	Live
Notification history log	Live
Manager dashboard	Live
Offline support with automatic sync	Live
Role-based access control	Live
Notification action buttons (Assign / Busy / Dismiss)	Live
Copper inventory tracking	Live

The Road Ahead: AI and Predictive Maintenance

The current platform is a foundation. Its long-term value lies in what the structured, standardised data it collects makes possible.

Phase 1 — Predictive Maintenance Analytics

Every job card completed today becomes a data point in a growing maintenance history. Machine learning models applied to this history will identify:

- **Failure patterns** — which machines fail most often, at what intervals, and under what conditions
- **Early warning signals** — combinations of faults that historically precede major failures
- **Predictive alerts** — "Machine X is due for a bearing failure based on 18 months of history — schedule inspection in the next 2 weeks"
- **Cost visibility** — which assets cost the most in repair time and parts, informing replacement or overhaul decisions

This shifts the maintenance function from **reactive** (fixing what breaks) to **predictive** (preventing it from breaking), directly reducing unplanned downtime and its associated production cost.

Phase 2 — AI Assistant for Technicians and Operators

A conversational AI assistant will be integrated into the app with access to:

- All operator and maintenance manuals for plant equipment
- Historical fault and repair records from job cards
- Wiring diagrams, parts lists, and technical schematics

A technician on the floor will be able to ask:

"What are the most common causes of overload faults on this conveyor?" "What was done the last three times this machine had a similar fault?" "What is the correct torque for the coupling on this gearbox?"

The assistant responds instantly with answers drawn from official documentation and real maintenance history — reducing diagnostic time and reducing dependency on senior staff for routine knowledge retrieval.

Business Case for AI Integration

Benefit	Mechanism
Reduced unplanned downtime	Predictive alerts trigger maintenance before failure
Faster fault resolution	AI-assisted diagnosis reduces time to identify root cause
Knowledge retention	Institutional knowledge captured in the system — not lost when experienced staff leave
Lower parts inventory cost	Predictive maintenance enables targeted stocking
Reduced repeat failures	Root cause captured and addressed, not just symptoms
Operator and technician upskilling	Access to manuals and history in seconds, on the phone

What Makes This Possible

The accuracy of AI-driven insights depends entirely on the quality of the data feeding them. Every job card completed with accurate machine identification, clear fault descriptions, and detailed closure notes becomes a data point that improves predictive accuracy.

This is why **data quality enforcement at the manager level is not an administrative task — it is a direct investment in the long-term intelligence of the maintenance operation.**

The structure being built today determines what the system can deliver in two years.

Summary

The CTP Job Cards system provides the business with:

1. **Speed** — critical faults reach the right technician in seconds, with automatic escalation to management if not responded to
2. **Accountability** — every fault is recorded, every response is timestamped, every action is logged
3. **Visibility** — management sees everything in real time without needing to chase information
4. **Compliance** — a complete, permanent audit trail for every maintenance event

5. **Intelligence** — a growing data asset that will power predictive maintenance and AI-assisted operations

The return on investment is measured in reduced unplanned downtime, lower mean-time-to-repair, and the progressive shift from emergency maintenance to planned, predictive maintenance — all of which translate directly to production output and cost efficiency.